Automobile Engineering

CSE - Semester 1 - 16 August 2026

Module 1: Electrical and Electronics System

Fundamentals of Engine Electricals

The engine electrical system provides power for starting, ignition, and charging. Key components include the battery (stores energy), starter motor (cranks engine), alternator (generates electricity to recharge battery and power systems), and ignition system (creates spark for combustion). Wiring harnesses connect these components, ensuring proper current flow and communication.

Without a functional engine electrical system, the vehicle cannot start, run, or maintain its battery charge. It's fundamental for engine operation and overall vehicle reliability.

Battery - Starter Motor - Alternator - Ignition System

Lighting and Indicators: Features & Requirements

Vehicle lighting includes headlights (low/high beam), tail lights, brake lights, turn signals, and interior lights. Requirements focus on visibility (to see and be seen), safety (indicating intentions), and legal compliance (specific brightness, color, and placement regulations). Modern systems often use LED technology for efficiency and longevity.

Proper lighting is critical for safe driving, especially at night or in adverse weather, by ensuring the driver can see the road and other road users can see the vehicle and its intended actions.

Headlights - Tail Lights - Turn Signals - LED Lighting

Body Electrical and Electronic Systems

These systems manage non-engine electrical functions, enhancing comfort, convenience, and safety. Examples include power windows, central locking, climate control, infotainment systems, and airbags. They often rely on electronic control units (ECUs) and communication networks (like CAN bus) to coordinate various components and sensors.

These systems significantly improve the user experience, comfort, and passive safety of a vehicle, making modern cars more appealing and functional beyond basic transportation.

ECU - CAN Bus - Infotainment - Central Locking

Monitoring and Instrumentation

This involves systems that provide drivers with crucial information about the vehicle's status and performance. The instrument cluster displays speed, RPM, fuel level, and warning lights. Electronic sensors monitor various parameters (e.g., engine temperature, oil pressure, tire pressure) and relay data to the

ECU and driver interface.

Monitoring and instrumentation allow drivers to operate the vehicle safely and efficiently, detect potential problems early, and comply with regulations by providing real-time operational data.

Instrument Cluster - Sensors - Warning Lights - Diagnostic Systems

Traditional vs. LED Vehicle Lighting

	Technology	Energy Efficiency	Lifespan	Response Time
Traditional (Halogen/Incandescen	Filament-based, heat and light from resistance.	Lower efficiency, significant heat loss.	Shorter, typically 500-2000 hours.	Slight delay, filament needs to heat up.
LED (Light Emitting Diode)	Semiconductor-based, light from electron recombination.	Higher efficiency, less heat, lower power draw.	Much longer, typically 15,000-50,000 hours.	Instantaneous, no warm-up time.

Module 2: Steering System

Camber

Camber is the inward or outward tilt of the wheel when viewed from the front of the vehicle. Positive camber means the top of the wheel tilts outwards, while negative camber means the top tilts inwards. It's measured in degrees.

Correct camber improves tire contact with the road, enhancing grip, stability, and steering response. Incorrect camber leads to uneven tire wear, reduced traction, and poor handling.

caster - king pin inclination - toe-in/out

Caster

Caster is the forward or backward tilt of the steering axis when viewed from the side of the vehicle. Positive caster means the top of the steering axis is tilted rearward, while negative caster means it's tilted forward.

Positive caster provides directional stability, helping the wheels return to the straight-ahead position after a turn. It also improves steering feel and high-speed stability, like a shopping cart wheel.

camber - king pin inclination - steering geometry

King Pin Inclination (KPI) / Steering Axis Inclination (SAI)

KPI is the inward tilt of the steering axis when viewed from the front of the vehicle. It's the angle between the steering axis and the vertical line through the wheel center.

KPI reduces steering effort, provides steering self-centering, and minimizes tire scrub during turns. It also creates a 'scrub radius' which influences steering stability and feedback.

camber - caster - scrub radius

Toe-in and Toe-out

Toe is the difference in distance between the front and rear of the tires when measured across the wheel. Toe-in means the front of the wheels are closer together than the rear. Toe-out means the front are further apart.

Toe settings compensate for forces acting on the wheels while driving, ensuring optimal tire contact and reducing tire wear. It affects straight-line stability and cornering response.

camber - caster - steering geometry

Understeering vs Oversteering

	Definition	Vehicle Behavior	Correction
Understeering	Loss of front wheel grip, vehicle turns less than intended.	Vehicle pushes wide in a turn, 'plowing' straight ahead.	Reduce throttle, lightly brake, reduce steering input.
Oversteering	Loss of rear wheel grip, vehicle turns more than intended.	Rear end slides out, vehicle wants to spin.	Counter-steer into the slide, modulate throttle.

Steering Geometry

Steering geometry refers to the precise angles and relationships of the steering and suspension components. It includes camber, caster, king pin inclination, and toe settings, all designed to work together.

Proper steering geometry is crucial for safe and predictable vehicle handling, tire longevity, and driver comfort. It ensures accurate steering response and stability at all speeds.

camber - caster - toe-in/out - Ackermann steering

Ackermann Steering Mechanism

The Ackermann steering mechanism ensures that during a turn, all wheels roll without slipping. The inner wheel turns at a sharper angle than the outer wheel, so their turning centers coincide.

> The Ackermann principle states that the axes of all wheels should intersect at a common point on the extension of the rear axle during a turn.

This mechanism minimizes tire scrub and wear during turns, improving steering efficiency and reducing stress on steering components. It's fundamental for smooth cornering.

steering geometry - toe-out on turn

Steering Ratio

$$R_s = \frac{\text{Steering Wheel Angle}}{\text{Road Wheel Angle}}$$

Symbol	Meaning
R_s	Steering Ratio
Steering Wheel Angle	Total rotation of the steering wheel
Road Wheel Angle	Average angle of the steered road wheels

Example

A car's steering wheel turns 3 full rotations (1080 degrees) from lock-to-lock. The road wheels turn 30 degrees from center to full lock. Calculate the steering ratio.

First, find the total road wheel angle from lock-to-lock: $30 \text{ degrees} \times 2 = 60 \text{ degrees}$.

Then, apply the formula: $R_s = \frac{1080 \text{ degrees}}{60 \text{ degrees}} = 18$.

The steering ratio is 18:1.

Applies when:

Typically measured from lock-to-lock or for a specific turn.

Module 3: Suspension System

Suspension System Principle

The suspension system connects the vehicle's wheels to its body, allowing relative motion. Its primary role is to absorb road shocks and vibrations, providing a smooth ride for occupants and protecting the cargo. It also maintains tire contact with the road surface for optimal traction and control during acceleration, braking, and cornering.

A well-designed suspension system is crucial for vehicle safety, stability, handling, and passenger comfort. It directly impacts tire wear and the longevity of other vehicle components.

ride comfort - handling - road holding - damping

Shock Absorbers

Shock absorbers, or dampers, are hydraulic devices that control the oscillations of the vehicle's springs. When a spring compresses or extends due to road irregularities, the shock absorber converts the kinetic energy of the spring's movement into heat energy, which is then dissipated. This prevents continuous bouncing and helps the vehicle quickly regain stability.

Without shock absorbers, springs would oscillate uncontrollably after hitting bumps, leading to a very uncomfortable ride, poor handling, and reduced tire contact with the road, compromising safety.

dampers - springs - damping force - oscillation control

Conventional vs Independent Suspension

	Wheel movement	Ride comfort	Handling	Unsprung mass
Conventional (Dependent)	Both wheels on an axle move together	Lower on rough roads	Less precise control	Higher
Independent	Each wheel moves independently	Higher, especially on uneven surfaces	Better, as each wheel reacts separately	Lower

Spring Types in Suspension

	Material/Mechanism	Characteristics	Common Use
Coil Spring	Helical steel wire	Compact, good energy storage	Most modern passenger cars
Leaf Spring	Stacked steel leaves	Simple, robust, high load capacity	Heavy-duty vehicles, trucks
Torsion Bar	Twisted metal bar	Compact, adjustable ride height	Some SUVs, older cars
Air Spring	Air bladder	Adjustable stiffness and height	Luxury cars, buses, trucks

Automatic/Hydro Suspension System

Automatic or hydro-pneumatic suspension systems use a combination of fluid (oil) and gas (nitrogen) to provide springing and damping. A central pump maintains fluid pressure, allowing the system to automatically adjust vehicle ride height and stiffness based on road conditions, load, and driver input. Sensors monitor various parameters to optimize performance.

These systems offer superior ride comfort and handling by dynamically adapting to changing conditions, improving stability during cornering and braking, and maintaining a consistent ride height regardless of load.

hydro-pneumatic - active suspension - adaptive damping - ride height control

Module 4: Wheels and Tyres

Wheel Types

Automobile wheels are primarily categorized by their construction material and manufacturing process. Common types include steel wheels, alloy wheels (aluminum or magnesium), and composite wheels. Each type offers different trade-offs in terms of weight, strength, cost, and aesthetics.

Wheel choice impacts vehicle performance (unsprung mass), fuel efficiency, ride comfort, and safety. Understanding types helps in maintenance, replacement, and performance tuning.

steel wheel - alloy wheel - composite wheel

Tyre Types

Tyres are classified based on their construction (radial, bias-ply), tread pattern (all-season, summer, winter, off-road), and intended use (passenger car, light truck, performance). Radial tyres are standard due to their flexibility and contact patch.

Correct tyre type ensures optimal grip, handling, braking, and fuel efficiency for specific driving conditions and vehicle types. Mismatched tyres compromise safety and performance.

radial tyre - bias-ply tyre - all-season tyre - winter tyre

Tyre Tread

Tyre tread refers to the pattern of grooves and blocks on the outer surface of the tyre that contacts the road. Its primary functions are to provide grip, channel water away for hydroplaning resistance, and dissipate heat. Tread patterns vary significantly for different driving conditions.

The tread is critical for traction, braking performance, and handling, especially in adverse weather. Worn tread significantly reduces safety and increases stopping distances.

tread pattern - grooves - sipes - hydroplaning

Tyre Selection Factors

Tyre selection involves considering vehicle specifications (size, load rating, speed rating), driving conditions (climate, terrain), and personal preferences (performance, longevity, noise). Manufacturer recommendations are crucial.

Choosing the right tyre ensures optimal vehicle performance, safety, fuel economy, and ride comfort. Incorrect selection can lead to premature wear, poor handling, and increased accident risk.

load rating - speed rating - tyre size - OE tyres

Radial vs Bias-Ply Tyres

	Construction	Flexibility	Contact Patch	Heat Generation	Common Use
Radial Tyre	Cords run radially (90 deg to tread)	High sidewall flexibility	Large, stable contact patch	Lower heat generation	Passenger cars, light trucks
Bias-Ply Tyre	Cords run diagonally (30-45 deg to tread)	Lower sidewall flexibility	Smaller, less stable contact patch	Higher heat generation	Trailers, older vehicles, some motorcycles